SANDEEP RATNANI

Senior Lead Software Engineer | Distributed Systems | AI/ML | Cloud & Event-Driven Architecture

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer with 11+ years of experience architecting and delivering production-grade distributed systems, cloud-native microservices, and AI-powered enterprise solutions. Deep expertise in Python, Java, Spring Boot, and AWS, with a strong track record of building high-throughput event-driven architectures processing 1M+ events/day. Skilled in designing LLM-powered AI agents, RAG pipelines, and real-time streaming systems deployed across 50+ manufacturing plants with 99.99% uptime. Proven ability to lead cross-functional engineering teams, mentor developers, and drive measurable impact in cost optimization, operational efficiency, and system reliability.

CORE SKILLS / TECHNICAL SKILLS

Programming Languages & frameworks: Java, Python, JavaScript, Spring Boot, Spring Security, Spring JPA, FastAPI, React, Redux
Backend: Spring boot, Fast API
Cloud & Infrastructure: AWS (EC2, Lambda, S3, DynamoDB, IoT Core, Kinesis, SQS, SNS, API Gateway, ECS, AWS Step Functions), Docker, Kubernetes, Helm
AI/ML: LangChain, LLM Integration, AI Agents, RAG Pipelines, Vector Databases (ChromaDB), Embeddings, Pandas, NumPy
Event Streaming: Apache Kafka, Kafka Streams, Schema Registry, RabbitMQ, MQTT, AMQP
Databases: PostgreSQL, DynamoDB, InfluxDB, OpenSearch, ChromaDB
Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Unified Namespace (UNS), REST, WebSockets, GraphQL
DevOps & Observability: GitHub Actions, Azure DevOps, New Relic, Prometheus, Grafana, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Senior Lead Software Engineer | John Deere, Pune | Apr 2022 - Feb 2026

- Design and implement **end-to-end data pipelines** for ingesting, processing, and preparing large-scale datasets to support AI/ML and Generative AI workloads.
- **Architected and implemented a Unified Namespace (UNS) platform** for manufacturing plants, enabling a real-time digital thread across disparate systems and reducing root-cause investigation time by 40%.
- Developed enterprise **AI agents** using LangChain and Python (FastAPI), integrated with the Unified Namespace, enabling AI-driven predictive decision-making across factory operations.
- Designed and deployed a **RAG**-powered troubleshooting system that processes live machine error codes, performs semantic retrieval over machine manuals using **embeddings and vector stores** (ChromaDB), and delivers automated diagnostic emails with actionable next-step instructions.
- **Served as Technical Lead** for cross-functional engineering teams, guiding developers through architectural decisions, code standards, deployment practices, and end-to-end observability implementation.
- Partnered directly with manufacturing leaders and plant operations to convert real-world production challenges into scalable digital and AI-driven backend solutions.
- Designed and built the SAP APM Link **microservices** ecosystem (orchestrator, audit service, batch processor, real-time processing) using Spring Boot and Kafka, enabling scalable **event-driven** APM integrations across the enterprise.

- Engineered cloud-based factory **asset tracking, notification**, and **location telemetry microservices on AWS**, reduced operational delays by 30% and improved asset visibility.
- Developed high-throughput streaming microservices with **Kafka Streams**, ensuring at-least-once delivery guarantees and reliably processing **1M+ events/day per factory**.
- Built and managed **multi-tenant Apache Kafka** and RabbitMQ clusters deployed on edge **Kubernetes**, powering real-time data pipelines across 50+ production plants with 99.99% uptime.
- Managed end-to-end deployment of containerized microservices on edge Kubernetes clusters using **Helm charts**, enabling automated, repeatable, multi-environment rollouts across 50+ global manufacturing sites.
- Implemented centralized authentication and authorization using **Spring Security, Okta OAuth2, and JWT tokens**, establishing role-based API protection and unified identity management across all microservices.
- Engineered OEE (Overall Equipment Effectiveness), asset lifecycle, shift management, and machine monitoring microservices, directly boosting plant operational efficiency by 25%.
- Implemented CI/CD pipelines and monitoring using GitHub Actions, Prometheus, Grafana, and New Relic.

Full Stack Engineer | Deloitte, Bengaluru | Feb 2019 - Feb 2022

- Engineered end-to-end **AWS IoT solutions** for large-scale manufacturing clients (Smart Factory, Reynolds, Pactiv), building scalable device-data ingestion, processing, and monitoring services using Java, AWS IoT Core, Lambda, S3, DynamoDB, and Kinesis.
- Implemented backend architecture for a **Digital Twin Smart Factory solution**, enabling real-time data ingestion, asset monitoring, and synchronization between physical equipment and digital models.
- Developed high-performance middleware services using **Java and GraphQL for Vanguard's banking platform**, optimizing API aggregation and reducing data retrieval latency by 35%.
- Built full-stack analytics features for Caterpillar using Java (Spring Boot), Spring JPA, Spring Security, and React, enhancing data visualization workflows and accelerating business insights delivery across multiple business units.

Senior Java Developer | KPIT Technologies, Pune | Dec 2017 - Feb 2019

- Built PLM connectors including a generic BOM Comparator tool, streamlining engineering change validation and reducing review cycles by 20%.
- Integrated ALM and PLM systems with ThingWorx mashups for Magna International, enabling seamless cross-platform data synchronization.
- Developed custom IoT extensions using ThingWorx Java SDK, extending platform capabilities for industrial clients.

Java + IoT Developer | ITC Infotech, Pune | Aug 2016 - Oct 2017

- Built real-time IoT dashboards for factory telemetry monitoring using ThingWorx, providing plant managers with actionable operational insights.
- Implemented an ML-powered Thing Predictor in Java for anomaly detection, contributing to a 15% reduction in unplanned machine downtime.

SQL + Java Developer | Subex, Pune | 2014 - Feb 2016

- Automated reporting engines using Java, eliminating manual operations and reducing report generation effort by 80%.
- Wrote optimized SQL queries for complex analytical reporting, improving query performance and enabling faster business decision-making.

KEY PROJECTS

Unified Namespace (UNS) Data Platform — Real-Time Digital Thread for Smart Manufacturing

Architected and developed a large-scale UNS-based data platform enabling a real-time digital thread across 10+ manufacturing plants. Built backend services for **data ingestion, contextualization, governance, and schema**

evolution, handling telemetry from **10,000+ devices per factory**. Designed **Kafka Streams pipelines** for high-throughput, low-latency event processing and stateful transformations. Implemented **historic data APIs** over InfluxDB for time-series queries and **schema discovery APIs** over OpenSearch for both current and historical device metadata. Deployed microservices on **Kubernetes**, ensuring horizontal scalability, multi-tenant isolation, and consistent observability. The platform unified OT/IT data, improved traceability, and **reduced investigation effort by 40%**.

AI-Driven Troubleshooting & Decision Intelligence Platform

Developed an enterprise-grade AI platform combining **RAG-powered diagnostics** and **LangChain-based intelligent agents** integrated with the Unified Namespace (UNS). Built pipelines that process **live machine error codes**, perform **semantic retrieval** over machine manuals using **LangChain + ChromaDB**, and generate automated remediation guidance. Additionally engineered **AI agents** exposed through **FastAPI** to support predictive, context-aware decision-making across factory operations—enabling autonomous insights and reducing troubleshooting time across plants.

SAP APM Link — Asynchronous, Event-Driven Microservices Platform

Designed an event-driven microservices ecosystem enabling real-time event injection and batch processing of device data for SAP Asset Performance Management. Built asynchronous services using **Spring Boot, Kafka, Spring Gateway, Discovery Service, and Spring Cloud Config**, with secure **JWT-based authentication** and **RBAC authorization**. Delivered a scalable integration layer deployed on edge Kubernetes for reliable, high-throughput SAP APM data flows.

Integrated OEE & Asset Intelligence Ecosystem

Architected and delivered a combined **OEE microservices suite** (Java, Spring Boot) and a cloud-native **Asset Track & Trace** platform built with Python on AWS ECS and a React frontend. Unified performance analytics and real-time asset tracking into a single operational intelligence layer, improving factory visibility and enabling data-driven decisions across manufacturing operations.

EDUCATION

Bachelor of Engineering in Electronics and Telecommunication - Sinhgad Academy of Engineering, Pune | 2014

Post Graduate Diploma - CDAC (Centre for Development of Advanced Computing) – ACTS, Aundh

CERTIFICATIONS

- AWS Certified Data Engineer – Associate
- AWS Certified Solutions Architect – Associate
- ThingWorx Developer Associate (DA)